

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 0.4522476354600182, median 0.36555257019483045, std: 0.3475573916591982
Gyroscope error (imu0): mean 1.0130172606876845e-07, median 1.2198256365022185e-10, std:
3.592103915796651e-06
Accelerometer error (imu0): mean 3.1173659532724863e-07, median 2.029305345359246e-08, std:
7.925072455517437e-06

Residuals

Reprojection error (cam0) [px]: mean 0.4522476354600182, median 0.36555257019483045, std: 0.3475573916591982
Gyroscope error (imu0) [rad/s]: mean 2.6790045405973954e-10, median 3.2259257030903225e-13, std:
9.499603880573745e-09
Accelerometer error (imu0) [m/s^2]: mean 8.200035450596436e-09, median 5.337960323382069e-10, std:
2.0846405605850904e-07

Transformation (cam0):

T_ci: (imu0 to cam0):

```
[ [ 0.99994815  0.00182842 -0.01001755  0.00002465]
  [ 0.00209833 -0.99963324  0.02699977  0.00021664]
  [-0.00996451 -0.02701939 -0.99958524  0.00001345]
  [ 0.          0.          0.          1.          ] ]
```

T_ic: (cam0 to imu0):

```
[ [ 0.99994815  0.00209833 -0.00996451 -0.00002497]
  [ 0.00182842 -0.99963324 -0.02701939  0.00021688]
  [-0.01001755  0.02699977 -0.99958524  0.00000784]
  [ 0.          0.          0.          1.          ] ]
```

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
-0.0004054542420946388

Gravity vector in target coords: [m/s^2]

[-0.00511757 -9.80648534 -0.03524061]

Calibration configuration

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cam0

Camera model: pinhole

Focal length: [489.32937658335146, 484.51791962688753]

Principal point: [393.69066058028494, 623.8932752948264]

Distortion model: radtan

Distortion coefficients: [-0.022394060339107905, -0.01032221722199354, -0.0016744900506054066, 0.002160133386853479]

Type: aprilgrid

Tags:

Rows: 6

Cols: 6

Size: 0.02 [m]

Spacing 0.006 [m]

IMU configuration

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IMU0:

Model: calibrated

Update rate: 200.0

Accelerometer:

Noise density: 0.00186

Noise density (discrete): 0.02630437226013957

Random walk: 0.000433

Gyroscope:

Noise density: 0.000187

Noise density (discrete): 0.0026445793616376874

Random walk: 2.66e-05

T_ib (imu0 to imu0)

[[1. 0. 0. 0.]

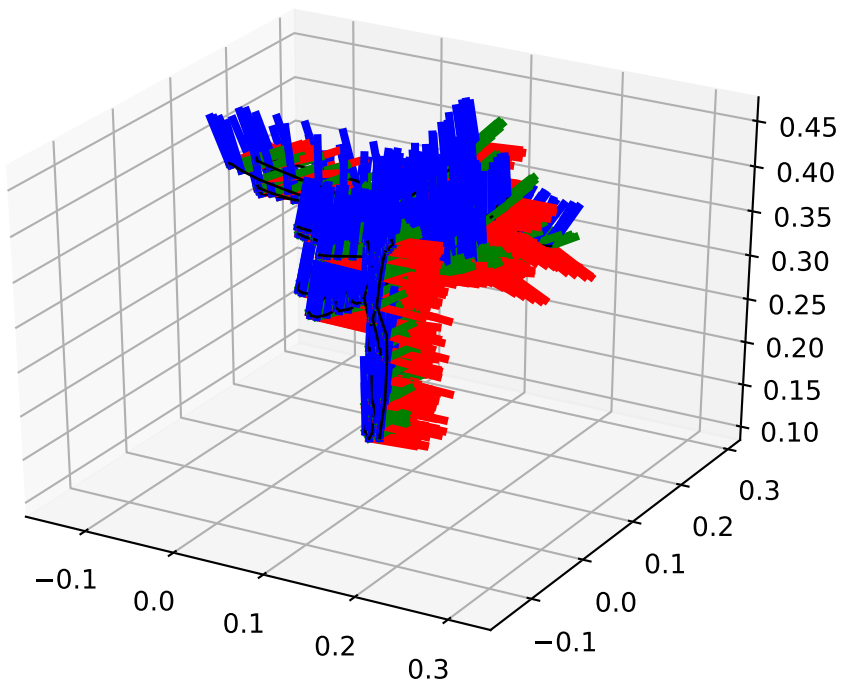
[0. 1. 0. 0.]

[0. 0. 1. 0.]

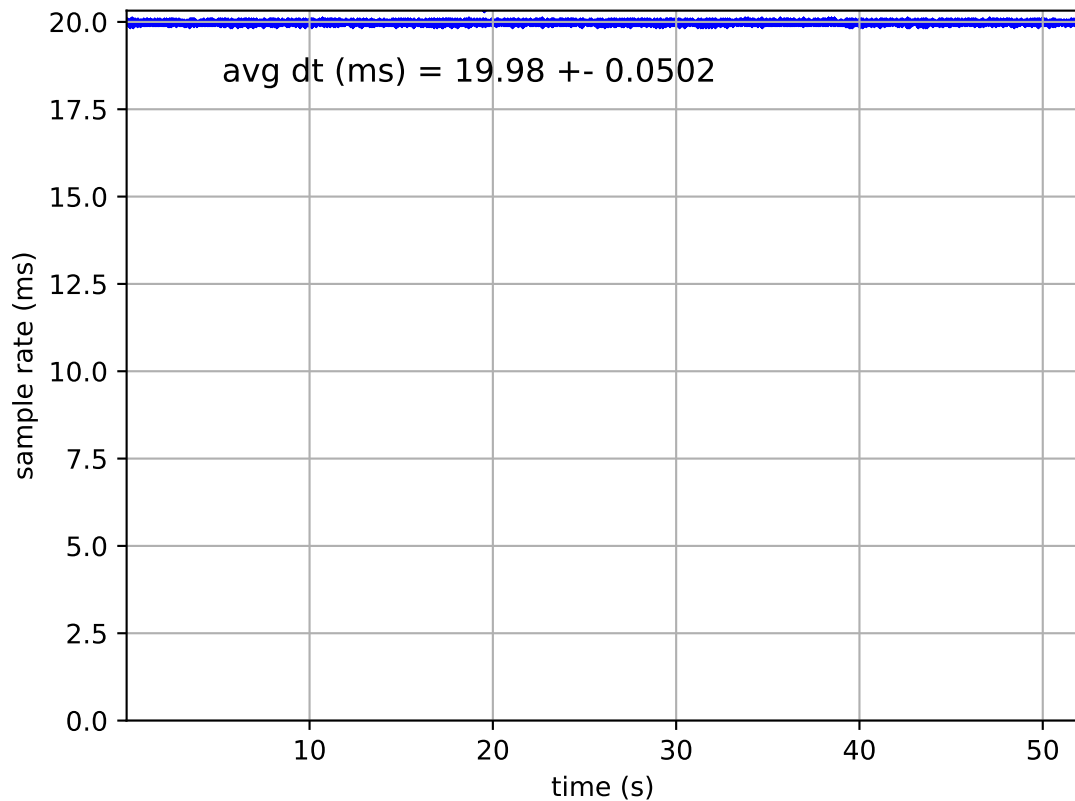
[0. 0. 0. 1.]]

time offset with respect to IMU0: 0.0 [s]

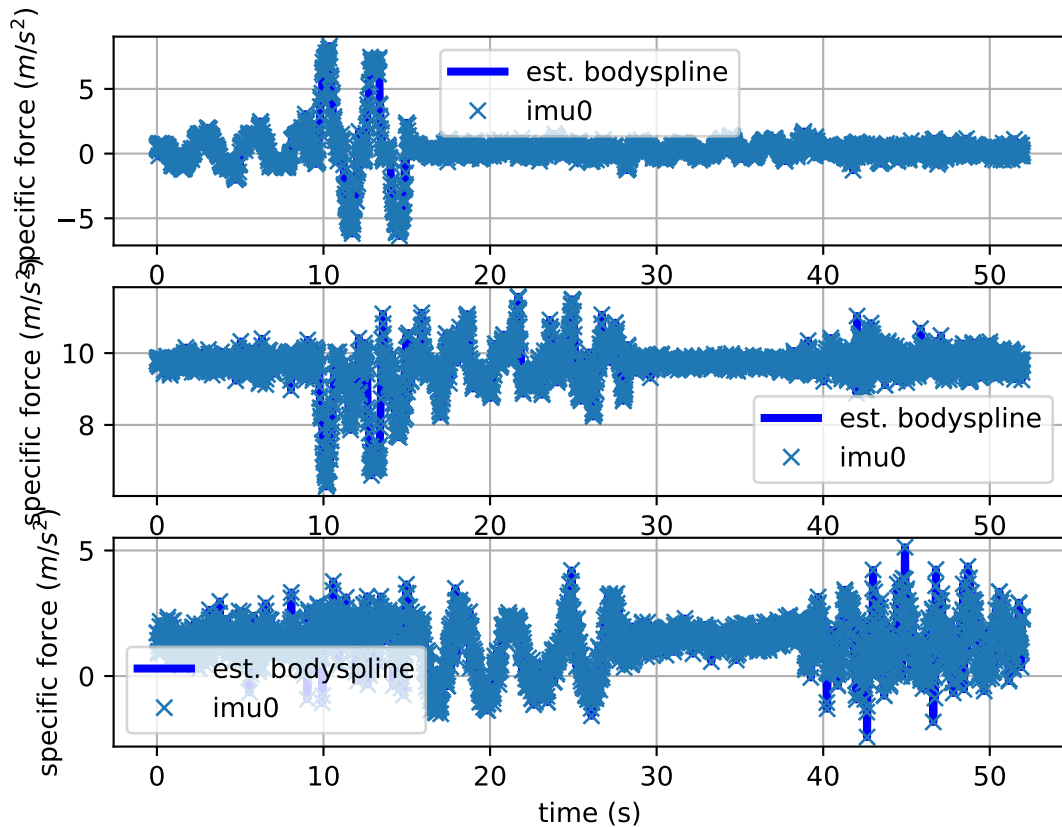
imu0: estimated poses



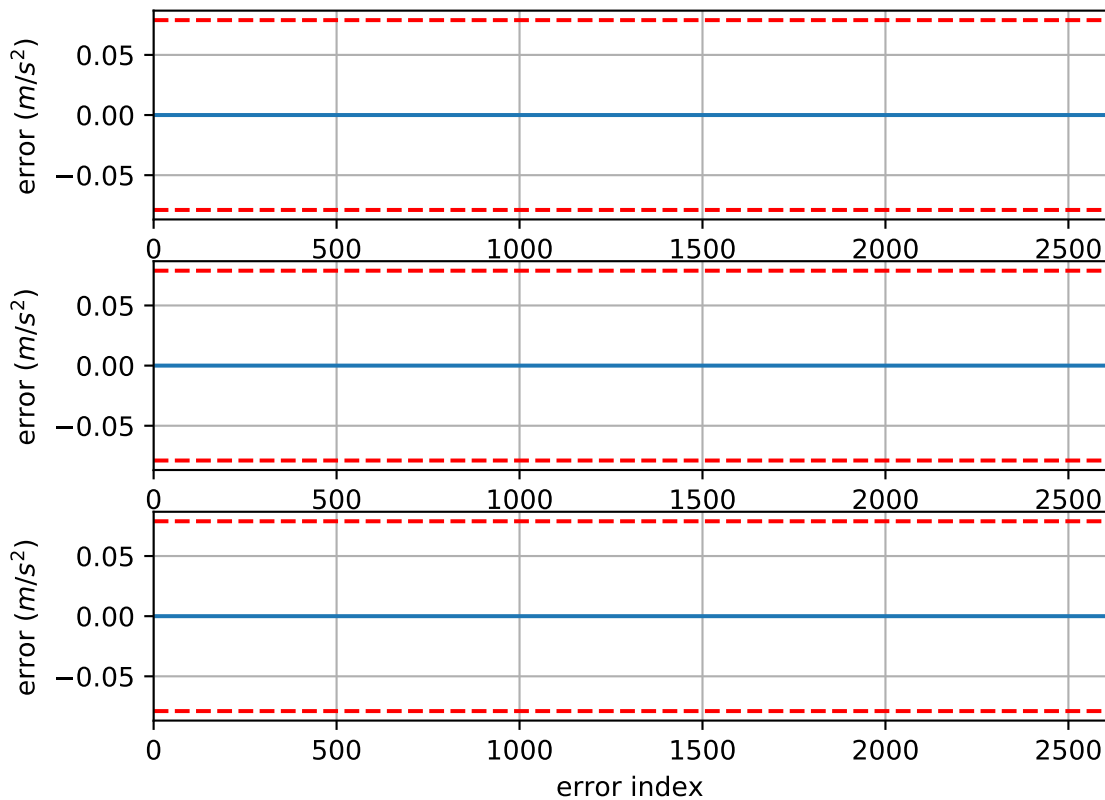
imu0: sample inertial rate



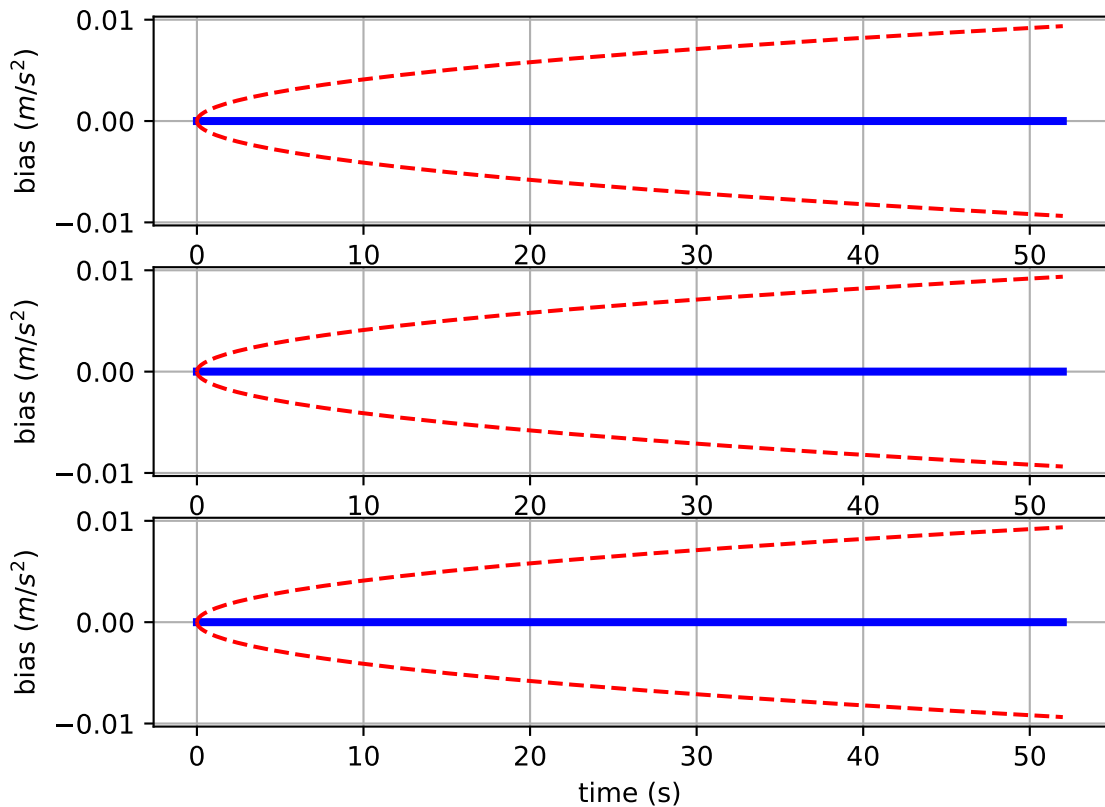
Comparison of predicted and measured specific force (imu0 frame)



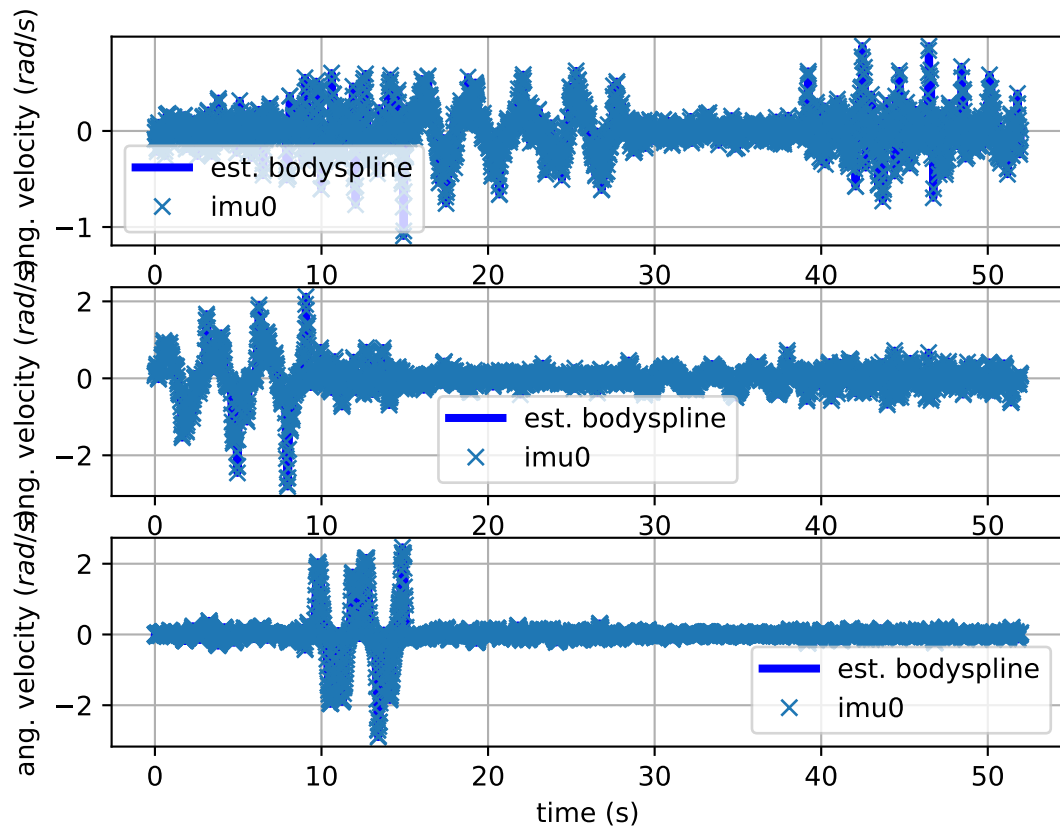
imu0: acceleration error



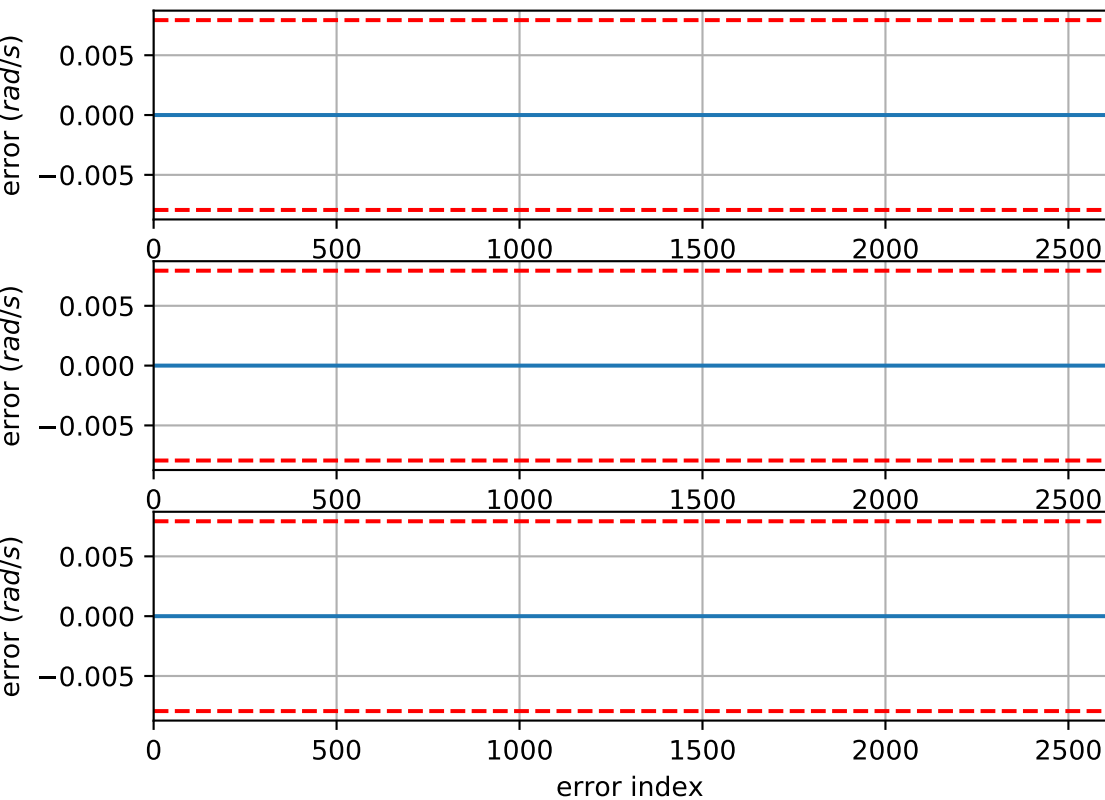
imu0: estimated accelerometer bias (imu frame)



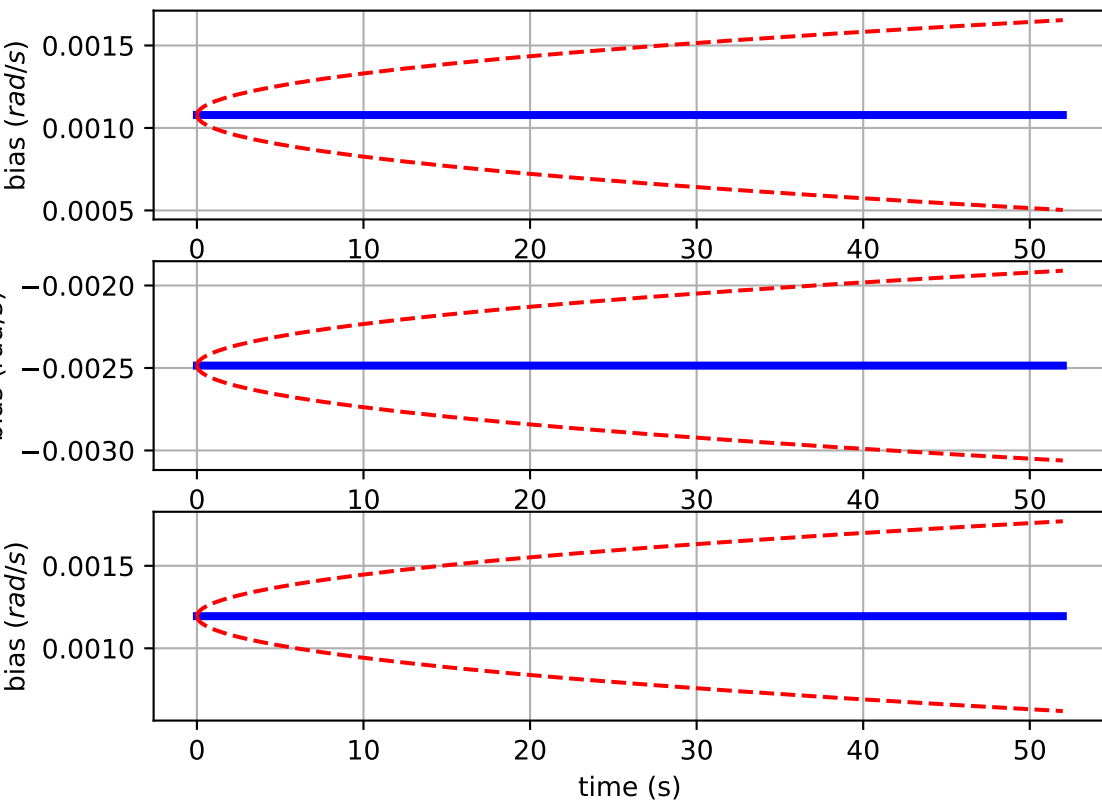
Comparison of predicted and measured angular velocities (body frame)



imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors

